



DPUHUDSGE

Simulates a simple head up display using the graphics engine SGE.

1. Inputs:

Show[1...10] Boolean values which indicate if the objects are displayed

2. Dynamic parameters:

x Position of all objects along the X axis
y Position of all objects along the Y axis
z Position of all objects along the Z axis (on-screen stacking). For HUD objects, $Z=-10..10$ is valid, and higher values are “on top” of lower values.
Scale Scaling factor for the objects

3. Static parameters:

Object[1...10] Names of the displayed objects
Visibility Visibility category for the objects. It can be used to display the objects on one of the views updated on a single graphics computer only.

4. Functional principle:

DPUHUDSGE can be used to simulate simple head up displays using the graphics engine SGE. To do so, one or multiple SGE objects are created which represent individual elements of the head up display.

Usually, objects will be displayed two-dimensionally over the simulation view. To this end, the object's *RenderClass* is set to *Overlay* and it is positioned in the 2D coordinate system.

More information about SGE objects is available in the manual *Reference Visualization SGE*.

When starting the simulation, DPUHUDSGE loads all objects that have been referenced in the parameters *Object1* to *Object10*. Object names must be specified in resource syntax; for example, “area2.bin3” corresponds to the SGE file “D:\SILAB\lib\graphics\default\models\area2\bin3.sig”. Resource naming is also detailed in the manual *Reference Visualization SGE*.

All loaded objects are displayed at the same on-screen location, which is set using the parameters *x* and *y*. Additionally, a scaling factor can be defined for all objects, which is set using the parameter *Scale*.

During runtime, the *i*'th object can be shown or hidden by modifying the input *Showi*.